Customer Segmentation in XYZ Bank using K-Means and K-Medoids Clustering

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***Abstract* — The Internet banking customers has growth very fast. Customer segmentation can be applied based on Internet banking data. Clustering is unsupervised data mining technique that can be used for customer segmentation. This research build clustering models on customer profile data based on their usage of Internet Banking in XYZ bank. The clustering methods employed K-Means method and K-Medoids method based on RFM score of customer's Internet Banking transactions. This research used Knowledge Discovery methodology [18]. The performances of both methods were measured and compared. The result shows that K-Means method outperformed K-Medoids method based on intra cluster (AWC) distance. While based on Davies-Bouldin index, K-Means performs slightly better than K-Medoids.**

Keywords — customer segmentation, k-means clustering, k-medoids clustering, data mining, bank

I. INTRODUCTION

The growth of Internet Banking customers is also followed by the increment of Internet Banking usages. The Internet Banking usages and customers from 2012 until 2014 were increased respectively about 1 trillion and 1 billion per year [1].

The objectives of data mining in banking sector are customer retention, fraud prevention and fraud detection [2]. Data mining was also used for customer segmentation [3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]. Customer segmentation will produce customer segments which each segment has similar characteristics [4].

If there are no predefined classes of data object, clustering can be used as unsupervised data mining technique [14]. Each cluster contains data points with high similarities but different significantly with data points of other clusters [4].

Many researches applied K-Means method for customer segmentation [4, 5, 6, 8, 9, 10, 11, 12, 13] and traffic management system [15].

In [14, 16] K-Means and K-Medoids method were used for text mining application to compare the performance of both methods in text clustering. K-medoids method does not employ centroid, but medoids [17]. K-Medoids outperformed K-Means clustering in text mining [16] and statistical clustering [18].

Customer segmentation and profiling employed RFM (Recency, Frequency, Monetary) method [6, 7, 8, 9, 10, 19] to know the loyalty of customer [7, 9]. While these researches [6, 8, 9, 11, 13] conducted customer segmentation to support CRM (Customer Relationship Management) in the company.

The goal of this research is to apply customer segmentation in XYZ bank using K-Means method and K-Medoids method based on their RFM score of Internet Banking transaction, in order to compare the performance of both clustering methods in customer segmentation.

II. LITERATURE REVIEW

K-Means Clustering was applied to build customer segmentation in mega business retail outfit for targeted customer services [4]. The background was because customers can be segmented based on demography and behaviors.

Customer segmentation was conducted using K-Means for milk industry to develop customer profile to explore their profitable [11] and market segmentation to learn customer's preferences [12].

These researches [5, 6] employed K-Means and decision tree as data mining approaches. K-Means was used to group bank customer based on their similarity in credit payment history [5] and RFM score [6]. The classification rules were developed using decision tree as classification technique to predict customer eligibility in credit approval [5] and categorize customer rank [6].

K-Means and RFM methods were used for customer segmentation [9, 10].

The researches proposed customer segmentation using RFM, K-Means and LEM2 methods in electronic company [8] and fertilizer manufacturer [13].

Compilation of customer segmentation researches is shown in table I.

TABLE I. COMPILATION OF CUSTOMER SEGMENTATION RESERACHES

Author	Year	Methods		
		K-Means	RFM	Optimum Number of Cluster
Ezenkwu, C.P., Ozuomba, S., and Kalu, C. [4]	2015	v		
Çaliş, A., Boyaci, A., and Baynal, K. [5]	2015	v		The smallest sum of squared error value

Khajvand, M., and Tarokh, M.J. [6]	2011	v	v	The Largest value of Dunn Index
Yuliari, N.P.P., Putra, I.K.G.D., and Rusjayanti, N.K.D. [7]	2015		v	MPC (Modified Partition Coefficient)
Cheng, C.H., and Chen, Y.S. [8]	2009	v	v	
Hosseini, S.M.S, Maleki, A., and Gholamian, M.R. [9]	2010	v	v	The smallest Davies-Bouldin Index value
Hamdi, K., and Zamiri, A. [10]	2016	v	v	
Lanjewar, R., and Yadav, O.P. [11]	2013	v		
Kashwan, K.R., and Velu, C. M. [12]	2013	v		
Dhandayudam, P., and Krishnamurthi, I. [13]	2013	v	v	

III. METHODOLOGY

A. Methods

This research used Knowledge Discovery methodology [20]:

Step 1: Data Cleaning

Table II shows attributes of Internet Banking Transaction. Meanwhile, data cleaning processes applied is shown in table III.

TABLE II. ATTRIBUTES OF INTERNET BANKING TRANSACTION

Attribute Name
TransactionDate
PostingDate
AccountNumber
CustomerID
DebitCreditCode
AmountOfTransaction
Balance

TABLE III. DATA CLEANING

Problem	Solution
There are non-numeric value in Balance and AmountOfTransaction attributes.	Deleted decimal value ".00", for example: 25000.00 was converted to 25000.

Step 2: Data Selection

The attributes that were selected are TransactionDate, AccountNumber, CustomerID, and AmountOfTransaction, in order to develop RFM method [6, 7, 8, 9, 10, 19] for customer segmentation.

Step 3: Data Transformation

RFM (Recency, Frequency and Monetary) Model

RFM model is used to learn the customer behavior [9, 20]. The variables used for RFM model are [6, 8, 9, 19, 20] are recency, frequency, and monetary.

We conducted data transformation using data aggregation which is grouped by CustomerID attribute and then quintiles approach [9, 19] to produce these variables:

1. Recency: interval between the last TransactionDate and the end of December 2015.
2. Frequency: how many transactions in one year (2015).
3. Monetary: total AmountOfTransaction in one year (2015).

Step 4: Data Mining

K-Means (KM) Clustering

In K-Means clustering, dataset, D, is given from n data points, and k, the number of cluster. Partitioning algorithm organize data points into partition (cluster), where $k \leq n$. [21].

The steps of K-means clustering are [21]:

1. Determine the number of cluster (k).
2. Select initial centroids.
3. Map each data point into a nearest cluster (most similar to centroid).
4. Update mean value (centroid) of each cluster.
5. Repeat step 3-4 until all centroids are not changed.

K-Medoids (K-Med) Clustering

K-Means minimize total squared error value, while K-Medoids minimize sum of dissimilarities between data points within cluster and select a data point in each cluster as center (medoids) [22]. Fig. 1 shows K-Medoids clustering algorithms.

Algorithm: *k*-medoids. PAM, a *k*-medoids algorithm for partitioning based on medoid or central objects.

Input:

- *k*: the number of clusters,
- *D*: a data set containing *n* objects.

Output: A set of *k* clusters.

Method:

(1) arbitrarily choose *k* objects in *D* as the initial representative objects or seeds;
(2) **repeat**
(3) assign each remaining object to the cluster with the nearest representative object;
(4) randomly select a nonrepresentative object, o_{random};
(5) compute the total cost, *S*, of swapping representative object, o_j, with o_{random};
(6) **if** $S < 0$ **then** swap o_j with o_{random} to form the new set of *k* representative objects;
(7) **until** no change;

Fig. 1. K-Medoids Clustering [21]

Step 5: Clustering Performance Calculation

Average Within Cluster (AWC)

The intra cluster (AWC) distance is calculated from the average of distance between a centroid and all cluster's elements (within cluster) [14].

Within-cluster or sum of variance/error (SSE) for each number of cluster, k is [21]:

$$\mathrm{SSE} = \sum \left(Y_i - \overline{Y_i} \right)^2 \quad (1)$$

where Y_i is value of each data points in cluster dan $\overline{Y_i}$ is centroid of cluster.

Davies-Bouldin Index (DBI)

DB index is calculated as follows [14]:

$$DB = \frac{1}{n} \sum_{i=1, i \neq j}^{n} max \left(\frac{\sigma_i + \sigma_j}{d(c_i, c_j))} \right) \quad (2)$$

where *n* is the number of clusters, *σi* is the intra cluster distance of cluster *i*, and *σj* is the intra cluster distance of cluster *j*, and *d*(*ci, cj*) is the distance between the centroid of cluster *ci* and *cj* .

The good clusters means that DBI value should be minimized [9, 14, 15].

B. Data

The data used in this research was collected from XYZ bank in Indonesia. We developed the dataset from customer's Internet Banking transactions from January 2015 until December 2015. There are 109,285 records of Internet Banking transaction and 3,043 active customers. Active customer is a customer who had at least 1 Internet Banking transaction in 2015.

The attributes that used for customer segmentation namely Recency (R), Frequency (F), and Monetary (M) [6, 7, 8, 9, 10, 19].

Table IV shows the attributes that were used for customer segmentation.

TABLE IV. ATTRIBUTES FOR CUSTOMER SEGMENTATION

Attribute Name	Data Range after Data Transformation
Recency (R)	65-363 days = 1 14-64 days = 2 9-13 days = 3 3-8 days = 4 0-2 days = 5
Frequency (F)	1 – 3 = 1 4 – 9 = 2 10 – 18 = 3 19 – 41 = 4 42 – 1,758 = 5
Monetary (M)	25,000 – 4,900,000 = 1 4,900,001 – 17,511,446 = 2 17,511,447 – 45,334,736 = 3 45,334,737 – 116,774,768 = 4 116,774,769 – 7,636,640,837 = 5

C. Research Framework

Fig. 2 shows the diagram block of research. After data collection, data preprocessing need to be conducted. Data preprocessing consists of data cleaning, data selection, and data transformation. Cluster model and two clustering performance methods was built and measured using Rapidminer 9.0 software.

This research applied two clustering techniques. They were K-Means and K-Medoids methods. The model was built using various k values (number of clusters) and Euclidean Distance as similarity measures. Intra cluster (AWC) distance and DBI were used to calculate the performance of clustering techniques.

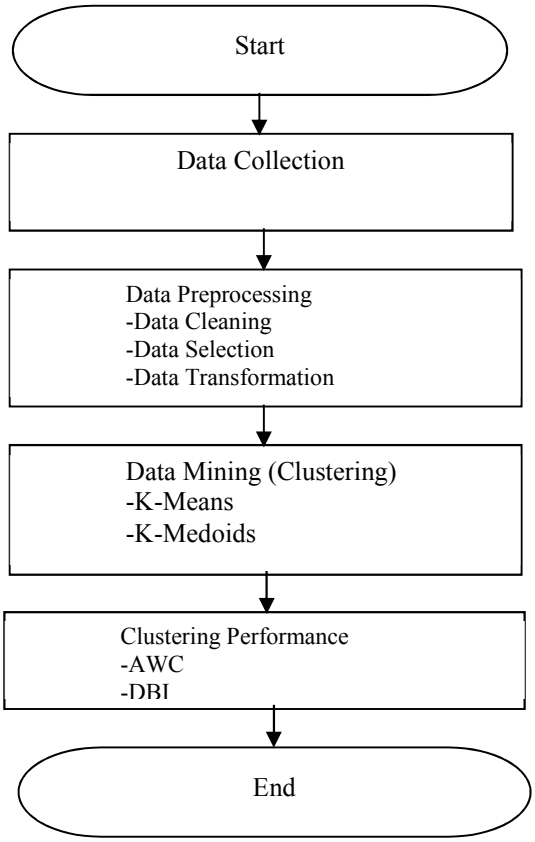


Fig. 2. Research Block Diagram

TABLE V. CLUSTERING PERFORMANCE

	AWC		DBI	
k	KM	K-Med	KM	K-Med
2	3.045	7.771	0.935	0.986
3	2.310	6.195	1.052	1.363
4	1.717	4.664	0.976	1.879
5	1.504	2.813	1.006	1.400
6	1.321	2.686	1.071	1.350
7	1.147	2.151	1.001	1.203
8	1.0267	1.839	0.941	1.736
9	0.959	1.687	0.982	1.329
10	0.941	1.527	-∞	1.366

IV. RESULTS

Table V shows the clustering performance. In term of AWC, the results show K-Means outperforms K-Medoids. This result did not support to these researches [16, 18].

While based on DBI, K-Means performs slightly better than K-Medoids for all cluster variety. Both clustering methods have better performance if DBI is used.

Based on the smallest value of DB index [9, 14, 15], the optimum number of cluster of both methods is 2.

TABLE VI. CLUSTER CHARACTERISTICS

RFM segments	**KM**		**K-Med**	
	Number of customers/ centroid	**Range of RFM score**	**Number of customers/ centroid**	**Range of RFM score**
Superstars -Highest F -Highest M	685/ R: 4.524 F:4.623 M: 4.442	12-15	589/ R: 2 F: 5 M: 4	7-14
Golden customers -High F -Second highest M	374/ R: 2.257 F: 4.377 M: 3.979	7-13	953/ R: 5 F: 4 M: 3	8-15
Typical customers -Average F -Average M	411/ R: 2.071 F: 2.336 M: 3.886	6-11	234/ R: 4 F: 1 M: 3	7-12
"Exceptional Occasions" customers -Second lowest F	448/ R: 3.395 F: 1.844 M: 1.819	5-9	213/ R: 4 F: 1 M: 1	5-9
"Everyday" Shoppers -Increased F - Medium to low M	466/ R: 4.455 F: 3.144 M: 2.758	9-12	30/ R: 5 F: 1 M: 2	8-9
Dormant customers -Low R -Low F -Low M	659/ R: 1.349 F: 1.340 M: 1.366	3-6	1024/ R: 2 F: 1 M: 2	3-8

We also built the cluster model based on RFM segments [23]. There are 6 customer categories, namely Superstars, Golden customers, Typical customers, Exceptional Occasions customers, Everyday shoppers, and Dormant customers. So we implemented cluster model with the number of cluster is 6. The characteristics of each cluster using K-means and K-Medoids is shown in table VI. RFM score is the summation of R, F, and M values with same weight (w=1).

The resultant clusters can be used to group customers based on their buying behavior. So the company can apply CRM, sales and marketing strategy to the targeted customers [6, 8, 9, 19].

V. CONCLUSIONS

This research built clustering models on customer profile data based on their usage of Internet Banking for customer segmentation in XYZ bank using K-Means method and K-Medoids method.

The performance of clustering result was measured and compared. K-Means method outperformed K-Medoids method based on intra cluster (AWC) distance. While based on DBI, K-Means performs slightly better than K-Medoids.

The optimum number of cluster of both methods is 2, based on the smallest value of DBI.

For future works: (1) clustering is applied by combining Internet Banking transaction data, Socio-demographic information and product ownership data, and (2) create rule of customer behavior using classification methods.

ACKNOWLEDGMENT

We gratefully acknowledge the supporting from RTTO (Research and Technology Transfer Office), Bina Nusantara University which helped the internal funding for this paper.

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